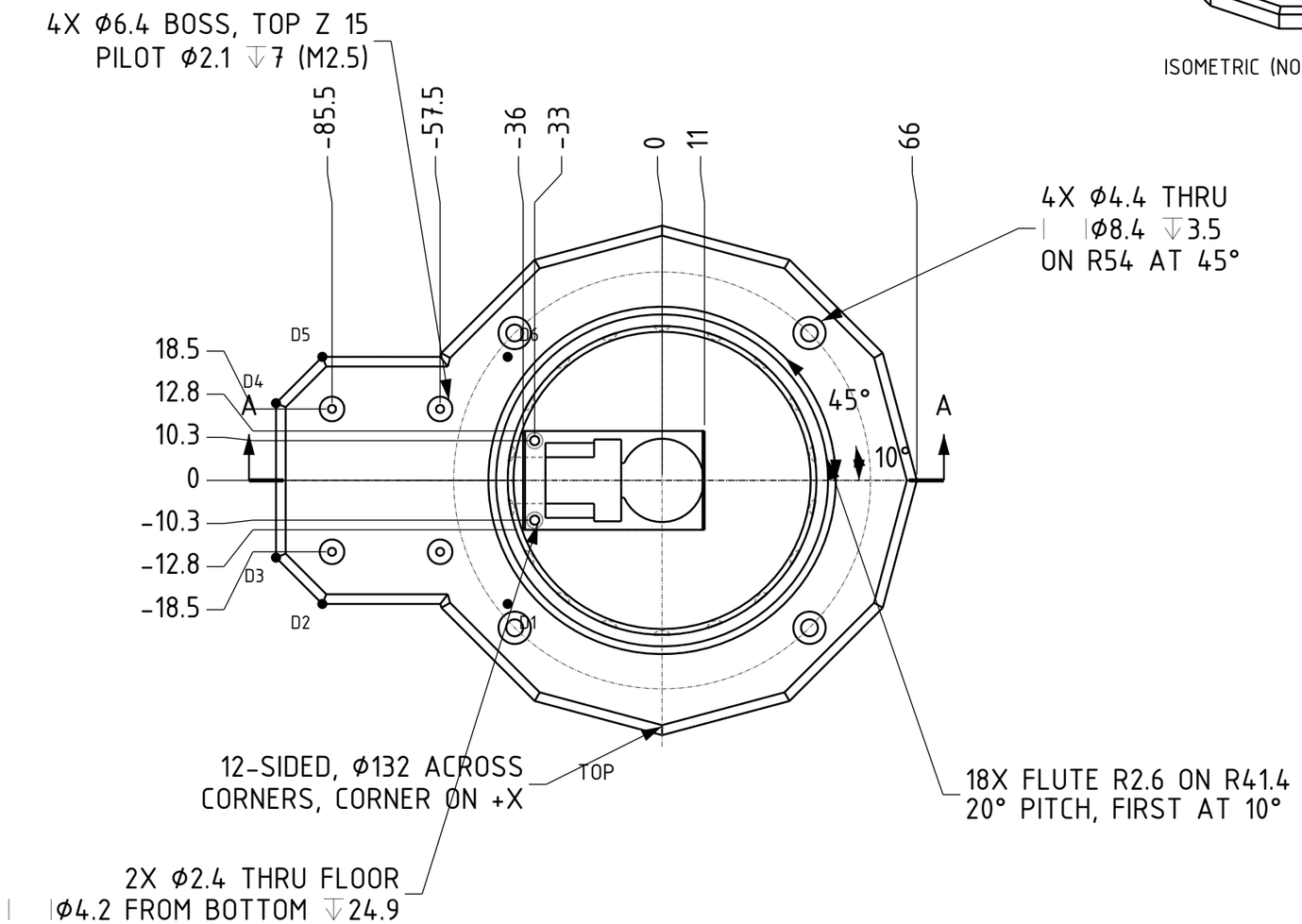
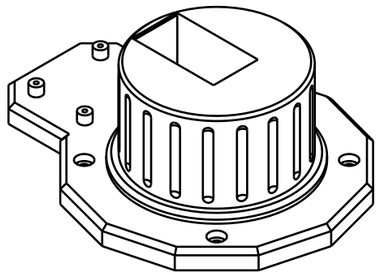




ISOMETRIC (NOT TO SCALE)

BOARD DECK POINTS (X, Y)

PT	X	Y
D1	-40	-32
D2	-88	-32
D3	-100	-20
D4	-100	20
D5	-88	32
D6	-40	32

FEATURES (in order)

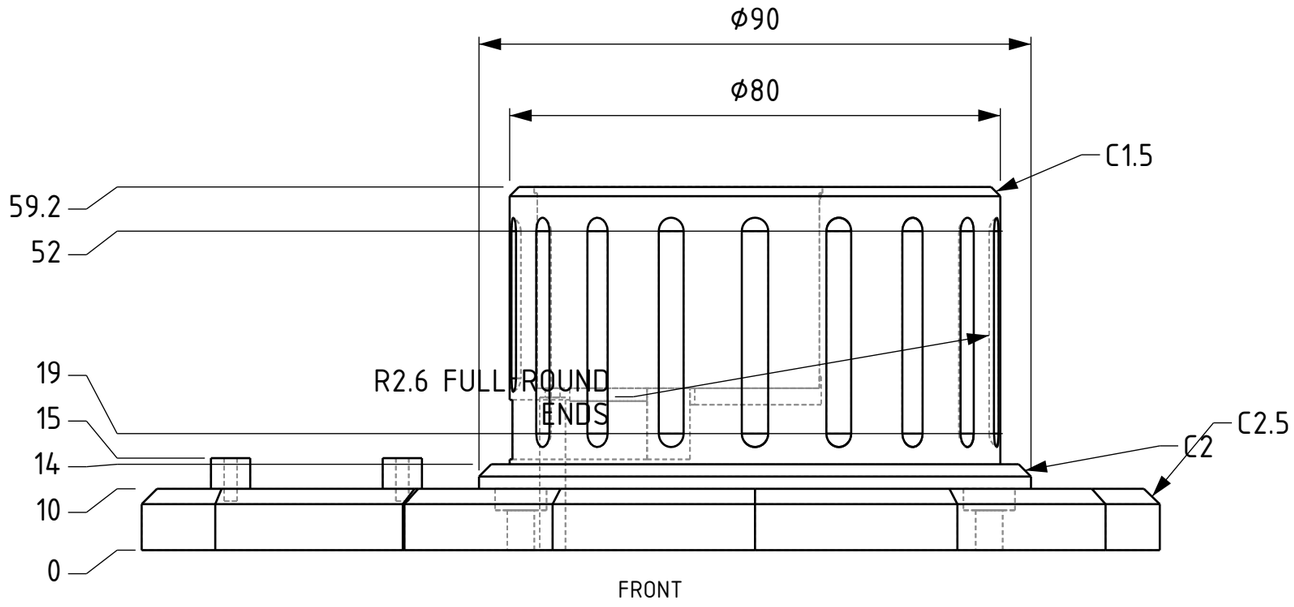
1. Plinth: regular 12-gon, ⌀132 across corners (a corner on +X), z 0 to 10.
2. Board deck: polygon D1 to D6, z 0 to 10; union with the plinth.
3. Chamfer C2.5 on all top outer edges (z 10).
4. Mounting holes: 4X ⌀4.4 through on R54 at 45° + 90°n; counterbore ⌀8.4 from z 6.5 up.
5. Collar ⌀90, z 10 to 14, C2 on its top edge.
6. Column ⌀80, z 10 to 59.2, C1.5 on its top edge.
7. Board standoffs: 4X ⌀6.4, z 9 to 15, at x -57.5 / -85.5, y ±18.5; pilot ⌀2.1 from z 8 to 16.
8. Flutes: 18X vertical ⌀5.2 on R41.4 at 10° + 20°n, axis z 19 to 52 (z 28 to 52 for the two at 170° and 190°, over the cable tunnel), spherical ends; cut.
9. Pocket P (sheet TAL-001) with its servo frame origin P0 at (0, 0, 43.3): servo +Z (shaft) along +Z, servo +X along +X, servo +Y along +Y.
10. Top opening: cut box x -36 to 11, y ±12.8, from z 58.2 up (the way in: the servo drops in from the top, connector end first).
11. Cable tunnel (plug bay to board deck): cut box x -60 to -13, y ±6, z 14.8 to 24.5. Pocket P's bump channel (z down to 24.3) opens into its roof.
12. Case screw holes from below (TAL-002, seat 18.4) at x -33, y ±10.3: ⌀2.4 from z 24.9 up to the pocket floor (z 26.4), ⌀4.2 counterbore from the bottom to z 24.9.

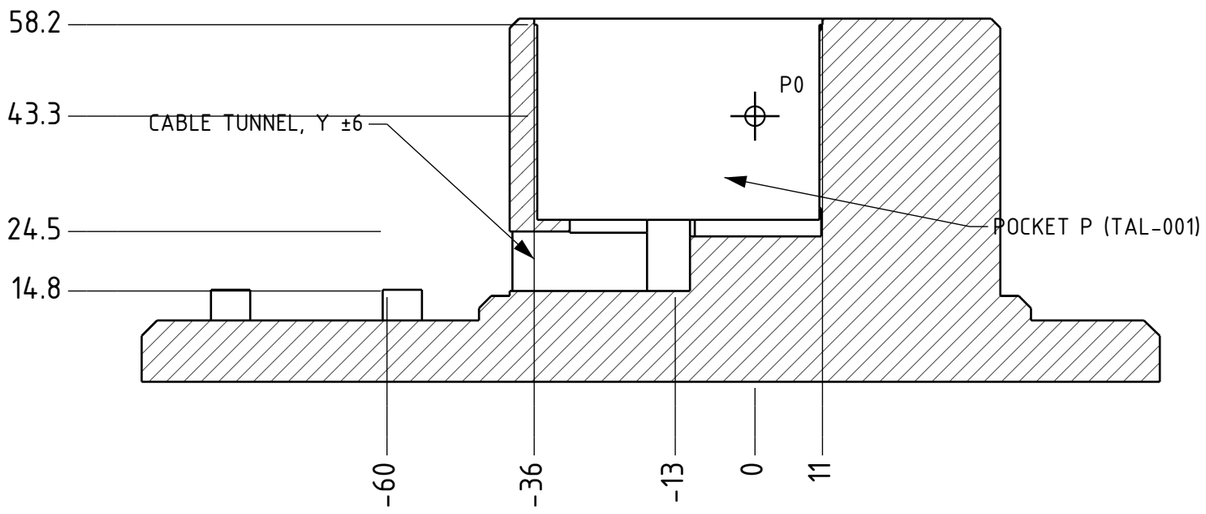
NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.
- The pan servo drops in from the top (its horns may stay on); its plugs hang into the bay under it. 2X M2 × 5 case screws from below.
- Print: plinth down; no support.

THIRD ANGLE PROJECTION

WORK SET A: base, shoulder, upper arm, hub caps, fit gauge	
PROJECT TALOS 5-axis arm	EST. MASS 154 g
PART Base (part sheet 1 of 2)	SCALE 2:3
PART NO. TAL-101	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-09
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 6 of 21





SECTION A-A

SECTION A-A

Section on the XZ plane (cutting line on sheet 1), looking toward +Y.

P0 = pocket P servo frame origin; servo axes along the part axes.

NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.

THIRD ANGLE PROJECTION

WORK SET A: base, shoulder, upper arm, hub caps, fit gauge	
PROJECT TALOS 5-axis arm	EST. MASS –
PART Base (part sheet 2 of 2)	SCALE 1:1
PART NO. TAL-101	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-09
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 7 of 21